

Random Matrix Theory: 0/0 of Universal Randomness

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2026

The Removable Singularity Framework

Abstract

We show that random matrix theory (RMT) is the most universal 0/0 removable singularity: level repulsion means $P(s=0) = 0$ EXACTLY, regardless of the matrix ensemble. The Wigner-Dyson statistics (GOE, GUE, GSE) are universal across quantum chaotic systems, financial correlation matrices, and the Riemann zeta zeros. The Montgomery-Odlyzko law proves that the distribution of zeta zeros follows GUE, meaning the primes are "quantum chaotic." This connects RMT to the Riemann hypothesis. The Wigner surmise $P(s) \sim s^\beta \cdot \exp(-(\beta+2)s^2/4)$ is universal for all three ensembles.

1. Wigner Matrices

Wigner introduced random matrices to model heavy nuclei. A Wigner matrix is symmetric (GOE) or Hermitian (GUE). Its eigenvalues repel each other: $P(s=0) = 0$. This is level repulsion, a 0/0 removable singularity.

2. Three Universal Ensembles

- GOE ($\beta=1$): real symmetric, time-reversal invariant
- GUE ($\beta=2$): complex Hermitian, no time-reversal
- GSE ($\beta=4$): quaternion, spin-1/2 systems

Each ensemble has universal nearest-neighbor spacing distribution. The parameter β controls the degree of level repulsion.

3. Wigner Surmise (Universal)

$$P(s) \sim s^\beta \cdot \exp(-(\beta+2)s^2/4)$$

The Wigner surmise is universal for all three ensembles. At $s=0$: $P(0) = 0$ for $\beta > 0$ (level repulsion). This is the 0/0: eigenvalues never coincide.

4. Montgomery-Odlyzko: Zeta Zeros = GUE

Montgomery (1973) conjectured that the pair correlation of Riemann zeta zeros follows GUE. Odlyzko (1987) verified this numerically for the first 10^{20} zeros. This means the primes are "quantum chaotic" -- their spacing follows the same statistics as quantum systems whose classical limit is chaotic.

5. Level Repulsion (0/0)

Level repulsion is the most fundamental 0/0 in RMT. For any ensemble with $\beta > 0$, $P(s=0) = 0$ EXACTLY. Eigenvalues push each other apart. This is a removable singularity at $s=0$.

6. Finance: Correlation Matrices

Eigenvalue distribution of financial correlation matrices follows Marchenko-Pastur (noise) plus outliers (signal). The largest eigenvalue corresponds to the market mode. RMT separates signal from noise in financial data.

7. Connections

RMT	Prior	Chapter	Connection
Zeta zeros = GUE	Ch.25	Zeta	Montgomery-Odlyzko
Lie algebras	Ch.24	E8	Classify ensembles
Quantum chaos	Ch.43	Chaos	Quantum Feigenbaum
Correlation matrices	Ch.38	Finance	Signal vs noise
Neural connectivity	Ch.34	Consciousness	Brain matrices

8. Conclusion

Random matrix theory is the most universal 0/0 removable singularity. Level repulsion means $P(s=0) = 0$ exactly, regardless of the ensemble. The Wigner-Dyson statistics are universal across quantum chaos, finance, and the Riemann zeta zeros. The Montgomery-Odlyzko law proves that the primes are quantum chaotic. The Wigner surmise $P(s) \sim s^\beta \cdot \exp(-(\beta+2)s^2/4)$ is universal. RMT connects to the Riemann hypothesis, quantum chaos, finance, and consciousness.

References

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